Learning and Rate-Adaptive Scheduling in Wireless Networks with Unknown Channel Statistics

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***Abstract*—Scheduling with unknown channel statistics couples online learning with queue stabilization, a challenge that arises in wireless access, service systems, etc. We study a single-server, multi-user queueing model in which the scheduler selects both a user and a service rate in each time slot. A transmission at the chosen (user, rate) succeeds with an unknown probability, and the outcomes are independent across time. Modeling each user–rate pair as an arm, we analyze UCB-informed rate-adaptation, which integrates MaxWeight scheduling with UCB exploration to learn the appropriate transmission rate for each user. We establish stability and performance bounds using Lyapunov drift analysis. Using synthetic simulations, we validate the algorithm's stability, showing that it closely tracks an oracle scheduler.**

***Index Terms*—UCB-informed rate-adaptation, Online Learning, Lyapunov Stability Analysis**

I. Introduction

Modern service systems increasingly require *rate-adaptive* scheduling decisions: the scheduler must choose *who* to serve and *at what service rate*, particularly when the service's success conditions are unknown. In wireless access networks (e.g., [1]), this corresponds to jointly scheduling a user and selecting a modulation-coding scheme (MCS), where higher rates offer larger payloads but also a higher probability of service failure when the instantaneous channel condition is weak. Analogous rate-adaptation phenomena arise in wireless networking systems (e.g., [2]–[6]), mmWave/beamformed links (e.g., [7]), and compute/storage systems that pick among multiple service configurations (e.g., [8]) that trade throughput for failure probability or latency. These applications motivate queueing models in which system stability and delay in service hinge on *joint* user-and-rate control rather than user selection alone (e.g., [1], [6], [7]).

We study a centralized single-server system with N queues in a slotted communication framework. In each slot, the scheduler selects a *user-rate* pair (u, m) where $u \in [1, \cdots, N]$ denotes a user and m denotes a service rate with a finite support: $m \in \{1, \ldots, M\}$. If a pair (u, m) is selected and succeeds, the service amount is $R_{u,m}$, which is a deterministic function of the selected (u, m) pair. If it fails, the service amount is 0. The channel statistics are unknown to the scheduler. A detailed discussion of the system model is provided in Section II.

When the channel statistics are known, MaxWeight scheduling is known to be throughput-optimal with near-optimal delay performance in several cases (see: [9]–[11]). In every slot t, MaxWeight policy for a joint user-rate scheduler chooses the (u, m) pair that maximizes the term $Q_u(t) \cdot \mu_{u,m}$ where $Q_u(t)$ is the queue length corresponding to user u and $\mu_{u,m}$ is the mean service rate for the pair (u, m). But the key difficulty in efficient scheduling in the setting when the channel statistics are unknown is that learning and queueing are tightly coupled: (i) the scheduler only observes outcomes for the selected pairs in the past, so inaccurate estimates for the channel statistics can persist for rarely chosen pairs; (ii) an efficient scheduling rule should depend on queue lengths, while queue lengths themselves evolve as a function of past (possibly misinformed) decisions; and (iii) in rate-adaptive wireless settings, outcomes across different m for the same user can be correlated within a slot through a single underlying channel state.

The above coupling between learning and user scheduling (between bullet (i) and (ii) above) is discussed in [12]. The authors in [12] argued through counterexamples that using a scheduler based on the sample mean of the service rates without using any optimism in the face of uncertainty results in queue instability.

To solve this coupling between learning and scheduling, recent works have used different strategies, as discussed in the following related work subsection.

A. Related Work

The authors in [13] use an online learning framework that decouples the learning and scheduling aspects in a frame-based approach, where they choose a set of queues in each slot for transmission, adhering to some interference constraints. In each frame that consists of multiple slots, channel statistics are learned anew while the queue lengths used for scheduling decisions are frozen at the beginning of each frame. Similar to [13], the authors in [14] consider a decentralized learning and scheduling setup where they considered an epoch-based approach where the learning part is executed for the users selected at the beginning of the learning phase (consisting of multiple slots) in each epoch. Simultaneous learning and scheduling in each slot was considered in [12] in their discounted UCB-MaxWeight ('UCB' stands for upper confidence bound) algorithm, where they considered a multi-server centralized setup to schedule different jobs with unknown service-time statistics. They consider a non-preemptive scheduling

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environment where a server is available only after it serves an assigned job. They also analyze a non-discounted version of their UCB-MaxWeight algorithm. While the analysis is comparatively simpler in a decoupled algorithm (e.g., [13]) than in [12], it results in inefficient learning as shown in the simulations in [12].

The above prior works do not consider a separate rate selection in addition to the user scheduling decision. Joint user scheduling and code-book selection (hence, the MCS selection) was studied in millimeter-wave networks in [7], where they propose one reinforcement learning (RL) based algorithm and one bandit-based approach. While this paper does not perform a theoretical analysis on the queue stability or the regret, they discuss the higher runtime of the RL-based approach compared to the bandit-based one due to the state-space in the corresponding MDP being large. It is worth mentioning that they do not use a UCB like optimistic approach to do the user scheduling; instead, they use the sample average of the channel statistics estimate to perform an empirical MaxWeight scheduling. In a similar motivation to ours, [1] studies joint user-rate selection with a focus on throughput regret and fairness violation over frames rather than physical packet-queue stability. At the beginning of each frame, they choose a feasible schedule and perform rate selection in a smaller time scale. It's worth mentioning that link rate selection using constrained Thompson sampling as the learning approach was considered in [6]. They provided upper and lower bounds on the expected regret. No queue stability was considered in the paper.

B. Contribution

Similar to [7], [12], our focus is on a centralized setting that isolates the learning and queue-scheduling coupling as the main technical obstacle. The main contributions of this paper are summarized below:

- We formalize a single-server, multi-user queueing model in which each action is a user-rate (u, m) pair and the channel statistics are unknown, capturing rate adaptation through correlated rate outcomes induced by different channel states for different users. We analyze a rate-adaptive UCB-informed rate-adaptation algorithm that treats each (u, m) pair as a separate arm while calculating the UCB bonus and solving the MaxWeight scheduling.
- For arrival rates strictly inside the capacity region with slack $\epsilon > 0$, we prove that the asymptotic average queue length is bounded by $O(1/\epsilon)$. The idea behind the corresponding proof is a decoupling Lemma 2, motivated by [14], that decouples the effect of learning and scheduling.
- We complement the theory with simulations under heterogeneous i.i.d. channels, demonstrating that the running time-average total queue length under the UCB-informed rate-adaptation algorithm asymptotically tracks an oracle scheduler that knows the channel statistics exactly.

While our paper is inspired by ideas in [12], [14], our model is quite different. In prior works, a packet is transmitted and takes a certain amount of time for the transmission to be completed based on channel conditions. In our model, motivated by [2], [4], [5], one can transmit at one of several rates and, success is guaranteed only if the unknown channel state can support the rate. However, unlike [2], [4], [5], we have queueing dynamics and our problem is not one of pure rate adaptation.

The rest of the paper is organized as follows: Section II presents the system model and capacity region and describes the MaxWeight-UCB algorithm in Algorithm II-B. Section III provides the main theoretical results and the corresponding proofs. Section V concludes.

II. System Model and the Algorithm

We consider a single-server multi-user slotted communication system with N users indexed by $u \in \{1, \dots, N\}$. Each slot is indexed by $t \geq 0$. Let $Q_u(t)$ denote the queue length of user u at the beginning of slot t. We consider i.i.d. arrivals across slots and users: say $A_u(t)$ packets arrive at queue u in slot t with mean $\lambda_u = \mathbb{E}[A_u(t)]$ and an upper bound $A_u(t) \leq A_{\max}, \forall u \in [1, \cdots, N]$. In each slot t, the scheduler selects exactly one user-rate pair (u, m), where $u \in \{1, \dots, N\}$ and $m \in \{1, \dots, M\}$.

We model the channel for each user u as follows: let $C_u(t)$ be the channel state for user u at time t and assume that the channel state can be in one of M different states, i.e., $C_u(t) \in \{1, \dots, M\}$. $C_u(t)$ is i.i.d. over time t. For a chosen (u, m) pair, if the rate $m \leq C_u(t)$, the transmission is successful; else, it fails. If the pair (u, m) is selected and succeeds, the service amount is $R_{u,m}$ packets, where $R_{u,m}$ is a deterministic bounded function of the pair (u, m). If it fails, the service amount is 0.

Denote by $(I(t), J(t))$ the user-rate pair selected at time t. For each user-rate pair (u, m), define the arm-level service random variable

$$S_{u,m}(t) := R_{u,m} \mathbf{1}\{C_u(t) \geq m\}.$$

Its mean is

$$\mu_{u,m} := \mathbb{E}[S_{u,m}(t)].$$

We assume $\mu_{u,m}$ is unknown to the scheduler. The realized service to queue u in slot t is

$$S_u(t) = \mathbf{1}\{I(t) = u\} \, S_{I(t),J(t)}(t).$$

Clearly,

$$0 \leq S_u(t) \leq S_{\max}, \qquad S_{\max} := \max_{u,m} R_{u,m}.$$

The queue evolution is

$$Q_u(t+1) = \max\{0, \ Q_u(t) + A_u(t) - S_u(t)\}. \tag{1}$$

A. Capacity region and performance objective

We assume the scheduler does *not* observe the instantaneous channel states $\{C_u(t)\}$ before choosing $(I(t), J(t))$; it only observes the outcome through the realized service.

a) Capacity region: We define the *capacity region* Λ as the set of arrival vectors $\lambda = (\lambda_1, \dots, \lambda_N)$ for which there exist nonnegative fractions $\{\alpha_{u,m}\}$ with $\sum_{u,m} \alpha_{u,m} = 1$ such that

$$\lambda_u \leq \sum_{m=1}^{M} \alpha_{u,m}\mu_{u,m}, \qquad \forall u \in \{1, \cdots, N\}. \tag{2}$$

We say that λ is *strictly* inside Λ with slack $\epsilon > 0$ if there exist such $\{\alpha_{u,m}\}$ for which

$$\lambda_u + \epsilon \leq \sum_{m=1}^{M} \alpha_{u,m}\mu_{u,m}, \quad \forall u \in \{1, \cdots, N\}.$$

b) Goal and metric: Our goal is to design an online policy (which does not know $\{\mu_{u,m}\}$) that stabilizes the system for every λ strictly inside Λ. Moreover, we seek an explicit bound on queueing delay via the time-average expected queue length:

$$\limsup_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} \sum_{u=1}^{N} \mathbb{E}[Q_u(t)].$$

B. Algorithm: UCB-informed rate-adaptation

Algorithm 1 UCB-informed rate-adaptation

1: **Initialization:** For all u, m, set $\hat{\mu}_{u,m}(0) = 0$ and $N_{u,m}(0) = 0$.
2: **for** $t = 0, 1, \dots$ **do**
3: **Update Indices:** For each pair (u, m), calculate the UCB: when $N_{u,m}(t) > 0$ (set it to ∞ when $N_{u,m}(t) = 0$):

$$b_{u,m}(t) = \sqrt{\frac{3S_{\max}^2 \ln(\max\{t, 2\})}{N_{u,m}(t)}}$$

$$\tilde{\mu}_{u,m}(t) = \min\left(S_{\max}, \hat{\mu}_{u,m}(t) + b_{u,m}(t)\right)$$

4: **Schedule:** Select the pair that maximizes the queue-weighted UCB:

$$(I(t), J(t)) \in \arg\max_{(u,m)} \left(Q_u(t) \cdot \tilde{\mu}_{u,m}(t)\right)$$

5: **Update Statistics:**
6: Observe service $S_{I(t),J(t)}(t)$.
7: Update counter: for each (u, m)

$$N_{u,m}(t+1) = N_{u,m}(t) + \mathbf{1}\{(I(t), J(t)) = (u, m)\},$$

8: Update empirical mean: for each (u, m)

$$\hat{\mu}_{u,m}(t+1) = \hat{\mu}_{u,m}(t) + \mathbf{1}\{(I(t), J(t)) = (u, m)\} \frac{S_{u,m}(t) - \hat{\mu}_{u,m}(t)}{N_{u,m}(t+1)}$$

9: **end for**

We treat each pair $k = (u, m)$ as a distinct arm. There are $K = N \times M$ total arms. The above UCB-informed rate-adaptation is considered in this paper. The policy couples learning and scheduling by combining a UCB estimate of the mean service $\mu_{u,m}$ with the queue length multiplier $Q_u(t)$, thereby biasing exploration toward actions that are most consequential for queue stability.

We note that given a channel condition $C_u(t)$ for a chosen user u, if a service rate m is successful, all rates below m would also be successful. This creates correlation for realized outcomes across rates at a fixed slot for a fixed user. However, for any fixed arm (u, m), the rewards observed at its pull times are i.i.d. because $C_u(t)$ is i.i.d. over time and the scheduler does not observe the current channel before acting. Hence, we can use standard UCB concentration ([15]) with some extra work to decouple the learning and scheduling, and we can treat each (u, m) as a separate arm in the UCB-informed rate-adaptation algorithm to stabilize the system for all arrival rates strictly inside the capacity region.

III. Main Results

This section states the stability guarantee for Algorithm II-B in Theorem 1 and provides a complete proof. Let $Q(t) := (Q_1(t), \dots, Q_N(t))$, $A(t) := (A_1(t), \dots, A_N(t))$, and $S(t) := (S_1(t), \dots, S_N(t))$ denote the queue-length, arrival, and service vectors at slot t, respectively. Throughout, let $\mathcal{H}_t$ denote the history generated by $\{Q(\tau), (I(\tau), J(\tau)), S(\tau), A(\tau)\}_{\tau=0}^{t-1}$, so that the scheduling decision at slot t is $\mathcal{H}_t$-measurable.

We assume the following slackness condition for the arrival process: there exist nonnegative fractions $\{\alpha_{u,m}\}$ with $\sum_{u,m} \alpha_{u,m} = 1$ such that for all $u \in [1, \cdots, N]$,

$$\lambda_u + \epsilon \leq \sum_{m=1}^{M} \alpha_{u,m}\mu_{u,m}. \tag{3}$$

Theorem 1 ($O(1/\epsilon)$ time-average expected total queue length)**.** *If λ satisfies* (3) *for some $\epsilon > 0$, then under Algorithm II-B, for every horizon $T \geq 2$,*

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\left[\sum_{u=1}^{N} Q_u(t)\right] \leq \frac{4B}{\epsilon} + \frac{4C_1(\ln T)^2}{\epsilon T} + \frac{4C_2}{\epsilon T} + \frac{4\,\mathbb{E}[L(0)]}{\epsilon T}, \tag{4}$$

where $L(0) := \frac{1}{2}\sum_{u=1}^{N} Q_u(0)^2$, and

$$B := \frac{N}{2}\left(A_{\max}^2 + S_{\max}^2\right), \tag{5}$$

and $C_1, C_2 < \infty$ are constants independent of T (given explicitly in the proof). Consequently,

$$\limsup_{T\to\infty} \frac{1}{T} \sum_{t=0}^{T-1} \sum_{u=1}^{N} \mathbb{E}[Q_u(t)] \leq \frac{4B}{\epsilon}. \tag{6}$$

A. Proof of Theorem 1

Before presenting the complete proof of Theorem 1, we provide a proof sketch here.

Proof Sketch:

1) We use $L(t) := \frac{1}{2}\sum_u Q_u(t)^2$ as a Lyapunov function, and analyze the Lyapunov drift conditioned on the filtration defined over the past. Using the slackness condition,

the Lyapunov drift, after summing over $t \in [0, T-1]$, decomposes into (i) a negative term $-\epsilon \sum_u Q_u(t)$ (ii) a regret-like term comparing the chosen action to the (unknown) MaxWeight action under true means.

2) Let $\mathcal{E}_t$ denote the "good" event that all empirical means are within their confidence intervals:
$$\mathcal{E}_t := \Big\{\forall (u,m):\ |\hat{\mu}_{u,m}(t) - \mu_{u,m}| \le b_{u,m}(t)\Big\}.$$
We decompose the regret term (ii), splitting it conditioned on the event $\mathcal{E}_t$ and $\mathcal{E}_t^c$, and we get:
   - Term I defined as
$$\sum_{t=0}^{T-1} \mathbb{E}\big[Q_{I(t)}(t) \min\{S_{\max}, 2b_{I(t),J(t)}(t)\}\big].$$
   - Term II defined as
$$S_{\max} \sum_{t=0}^{T-1} \mathbb{E}\Big[\Big(\sum_{u=1}^{N} Q_u(t)\Big)\mathbf{1}(\mathcal{E}_t^c)\Big].$$

3) The key challenge in bounding terms I and II is that the standard UCB concentration bounds (e.g., [15]) cannot be applied to these terms due to the coupling of the bonus or the bad event term with the queue length of the chosen arm, which can be arbitrarily large. We use a decoupling Lemma (Lemma 2), first established for a different model in [14], which decouples these terms into an average queue length term and an error term corresponding to the selection of sub-optimal arms. Standard UCB concentration bounds can be applied after this decoupling.

4) Finally, we telescope the drift to obtain the finite-time bound (4) and the asymptotic bound (6).

1) Lyapunov Drift Decomposition: Define the one-step conditional drift
$$\Delta(t) := \mathbb{E}[L(t+1) - L(t) \mid \mathcal{H}_t].$$
From the queue recursion $Q_u(t+1) = \max\{0, Q_u(t) + A_u(t) - S_u(t)\}$ and the inequality
$$\frac{1}{2}\big(\max\{0, x+a-s\}\big)^2 - \frac{1}{2}x^2 \le x(a-s) + \frac{1}{2}(a^2+s^2),$$
with $x = Q_u(t)$, $a = A_u(t)$, and $s = S_u(t)$, we obtain after summing over u:
$$\Delta(t) \le \sum_{u=1}^{N} Q_u(t)\, \mathbb{E}[A_u(t) - S_u(t) \mid \mathcal{H}_t] \tag{7}$$
$$+ \frac{1}{2}\sum_{u=1}^{N} \mathbb{E}[A_u(t)^2 + S_u(t)^2 \mid \mathcal{H}_t]$$
$$\le \sum_{u=1}^{N} Q_u(t)\lambda_u - \mathbb{E}\left[\sum_{u=1}^{N} Q_u(t) S_u(t) \,\middle|\, \mathcal{H}_t\right] + B. \tag{8}$$
Since exactly one pair $(I(t), J(t))$ is scheduled, $\sum_u Q_u(t)S_u(t) = Q_{I(t)}(t) S_{I(t),J(t)}(t)$ and $\mathbb{E}[S_{I(t),J(t)}(t) \mid \mathcal{H}_t] = \mu_{I(t),J(t)}$, hence
$$\Delta(t) \le B + \sum_{u=1}^{N} Q_u(t)\lambda_u - Q_{I(t)}(t)\mu_{I(t),J(t)}. \tag{9}$$
Multiply (3) by $Q_u(t)$ and sum over u:
$$\sum_{u=1}^{N} Q_u(t)\lambda_u \le \sum_{u=1}^{N}\sum_{m=1}^{M} \alpha_{u,m} Q_u(t)\mu_{u,m} - \epsilon \sum_{u=1}^{N} Q_u(t). \tag{10}$$
Let $(u^\star(t), m^\star(t)) \in \arg\max_{u,m} Q_u(t)\mu_{u,m}$ be a maximizer under the *true* means. Since $\alpha_{u,m} \ge 0$ and $\sum_{u,m} \alpha_{u,m} = 1$,
$$\sum_{u,m} \alpha_{u,m} Q_u(t)\mu_{u,m} \le Q_{u^\star(t)}(t)\mu_{u^\star(t),m^\star(t)}. \tag{11}$$
Combining (9)–(11) yields
$$\begin{aligned}\Delta(t) &\le B - \epsilon \sum_{u=1}^{N} Q_u(t) \\ &\quad + \underbrace{\Big(Q_{u^\star(t)}(t)\mu_{u^\star(t),m^\star(t)} - Q_{I(t)}(t)\mu_{I(t),J(t)}\Big)}_{=:\ Reg(t)}.\end{aligned} \tag{12}$$

2) Simplifying the Regret Term $Reg(t)$: Recall the UCB bonus radii $b_{u,m}(t)$ (for $t \ge 2$) defined in the Algorithm II-B. Let $\mathcal{E}_t$ denote the "good" event that all empirical means are within their confidence intervals:
$$\mathcal{E}_t := \Big\{\forall (u,m):\ |\hat{\mu}_{u,m}(t) - \mu_{u,m}| \le b_{u,m}(t)\Big\}.$$

Lemma 1 (UCB concentration)**.** *For all $t \ge 2$,*
$$\mathbb{P}(\mathcal{E}_t^c) \le 2K\, t^{-4}.$$

Proof. Fix an arm (u,m). Let $Y_{u,m}(s)$ denote the reward observed on the s-th pull of arm (u,m). Since the channel process is i.i.d. over time, the sequence $\{Y_{u,m}(s)\}_{s\ge1}$ is i.i.d. with mean $\mu_{u,m}$ and is bounded in $[0, S_{\max}]$. On the event $\{N_{u,m}(t) = n\}$ with $n \ge 1$, the empirical mean satisfies $\hat{\mu}_{u,m}(t) = \frac{1}{n}\sum_{s=1}^{n} Y_{u,m}(s)$. Therefore, by Hoeffding's inequality,
$$\begin{aligned}&\mathbb{P}\left(|\hat{\mu}_{u,m}(t) - \mu_{u,m}| > \sqrt{\frac{3S_{\max}^2 \ln t}{n}} \,\middle|\, N_{u,m}(t) = n\right) \\ &\le 2\exp(-6\ln t) = 2t^{-6} \le 2t^{-4}.\end{aligned}$$
Unconditioning and taking a union bound over all $K = NM$ arms yields
$$\mathbb{P}(\mathcal{E}_t^c) \le 2Kt^{-4}.$$
□

Now we analyze the term $Reg(t)$ on $\mathcal{E}_t$ and $\mathcal{E}_t^c$ separately. On $\mathcal{E}_t$, optimism implies $\mu_{u,m} \le \tilde{\mu}_{u,m}(t)$ for all (u,m), hence
$$\begin{aligned}Q_{u^\star(t)}(t)\mu_{u^\star(t),m^\star(t)} &\le Q_{u^\star(t)}(t)\tilde{\mu}_{u^\star(t),m^\star(t)}(t) \\ &\le Q_{I(t)}(t)\tilde{\mu}_{I(t),J(t)}(t),\end{aligned}$$
where the last inequality uses the argmax rule of Algorithm II-B. Therefore,
$$\begin{aligned}Reg(t)\mathbf{1}(\mathcal{E}_t) &\le Q_{I(t)}(t)\big(\tilde{\mu}_{I(t),J(t)}(t) - \mu_{I(t),J(t)}\big)\mathbf{1}(\mathcal{E}_t) \\ &\le Q_{I(t)}(t)\min\Big\{S_{\max},\ 2b_{I(t),J(t)}(t)\Big\},\end{aligned} \tag{13}$$

since on $\mathcal{E}_t$ we have $\tilde{\mu}_{I(t),J(t)}(t) = \min\{S_{\max},\, \hat{\mu}_{I(t),J(t)}(t) + b_{I(t),J(t)}(t)\} \le \min\{S_{\max},\, \mu_{I(t),J(t)} + 2b_{I(t),J(t)}(t)\}$. On $\mathcal{E}_t^c$, boundedness gives $\mu_{u,m} \le S_{\max}$ and thus

$$Reg(t)\mathbf{1}(\mathcal{E}_t^c) \le S_{\max}\Big(\sum_{u=1}^{N} Q_u(t)\Big)\mathbf{1}(\mathcal{E}_t^c). \tag{14}$$

Taking expectations in (12) and using (13)–(14) yields

$$\begin{aligned}\mathbb{E}[\Delta(t)] \le{}& -\epsilon\,\mathbb{E}\Big[\sum_{u=1}^{N} Q_u(t)\Big] + B \\ &+ \mathbb{E}\big[Q_{I(t)}(t)\min\{S_{\max},\, 2b_{I(t),J(t)}(t)\}\big] \\ &+ S_{\max}\,\mathbb{E}\Big[\Big(\sum_{u=1}^{N} Q_u(t)\Big)\mathbf{1}(\mathcal{E}_t^c)\Big].\end{aligned} \tag{15}$$

Summing (15) over $t = 0, \ldots, T-1$ and telescoping $\sum_{t=0}^{T-1}\mathbb{E}[\Delta(t)] = \mathbb{E}[L(T)] - \mathbb{E}[L(0)] \ge -\mathbb{E}[L(0)]$ gives

$$\epsilon\sum_{t=0}^{T-1}\mathbb{E}\Big[\sum_{u=1}^{N} Q_u(t)\Big] \tag{16}$$

$$\begin{aligned}&\le \underbrace{\sum_{t=0}^{T-1}\mathbb{E}\big[Q_{I(t)}(t)\min\{S_{\max},\, 2b_{I(t),J(t)}(t)\}\big]}_{I} \\ &\quad + \underbrace{S_{\max}\sum_{t=0}^{T-1}\mathbb{E}\Big[\Big(\sum_{u=1}^{N} Q_u(t)\Big)\mathbf{1}(\mathcal{E}_t^c)\Big]}_{II} + \mathbb{E}[L(0)] + BT.\end{aligned} \tag{17}$$

Next, we bound the terms I and II.

3) Bounding Term I and Term II: To handle the coupling between learning and scheduling, we use a decoupling idea from [14] tailored to our setting:

Let $Q_{\text{sum}}(t) := \sum_{u=1}^{N} Q_u(t)$ and define $D_{\max} := N\max\{A_{\max}, S_{\max}\}$. Then $|Q_{\text{sum}}(t+1) - Q_{\text{sum}}(t)| \le D_{\max}$ deterministically.

Lemma 2 (A decoupling lemma for queue-weighted rare events)**.** *Let $\{X_t\}_{t\ge 0}$ be nonnegative and satisfy $X_t \le X_{t-k} + kD_{\max}$ for all $t \ge 0$ and integers $k \ge 0$ (with $X_\tau = 0$ for $\tau < 0$). Let $\{E_t\}_{t=0}^{T-1}$ be some events and $N_E := \sum_{t=0}^{T-1}\mathbf{1}(E_t)$. Then for any $\delta > 0$,*

$$\begin{aligned}\sum_{t=0}^{T-1}\mathbb{E}\big[X_t\mathbf{1}(E_t)\big] \le{}& \delta\sum_{t=0}^{T-1}\mathbb{E}[X_t] \\ &+ \frac{D_{\max}}{2}\Big(\frac{1}{\delta} + 1\Big)\mathbb{E}[N_E^2].\end{aligned} \tag{18}$$

Proof. Fix a sample path (so that all quantities below are deterministic) and define the random window length

$$w := \left\lceil \frac{N_E}{\delta} \right\rceil \vee 1.$$

For any t and any $k \in \{0, 1, \ldots, w-1\}$, the bounded-increment property implies

$$X_t \le X_{t-k} + kD_{\max}.$$

Averaging over k yields

$$\begin{aligned}X_t &\le \frac{1}{w}\sum_{k=0}^{w-1} X_{t-k} + \frac{1}{w}\sum_{k=0}^{w-1} kD_{\max} \\ &= \frac{1}{w}\sum_{k=0}^{w-1} X_{t-k} + \frac{w-1}{2}D_{\max}.\end{aligned}$$

Multiply both sides by $\mathbf{1}(E_t)$ and sum over $t = 0, \ldots, T-1$:

$$\sum_{t=0}^{T-1} X_t\mathbf{1}(E_t) \le \frac{1}{w}\sum_{t=0}^{T-1}\mathbf{1}(E_t)\sum_{k=0}^{w-1} X_{t-k} \tag{19}$$

$$\begin{aligned}&+ \frac{w-1}{2}D_{\max}\sum_{t=0}^{T-1}\mathbf{1}(E_t) \\ &\le \frac{1}{w}\sum_{t=0}^{T-1}\mathbf{1}(E_t)\sum_{k=0}^{w-1} X_{t-k} + \frac{wD_{\max}}{2}N_E.\end{aligned} \tag{20}$$

We now rewrite the first term in (20). For each fixed t,

$$\sum_{k=0}^{w-1} X_{t-k} = \sum_{\tau=t-w+1}^{t} X_\tau,$$

where we use the convention $X_\tau = 0$ for $\tau < 0$. Hence

$$\sum_{t=0}^{T-1}\mathbf{1}(E_t)\sum_{k=0}^{w-1} X_{t-k} = \sum_{t=0}^{T-1}\mathbf{1}(E_t)\sum_{\tau=t-w+1}^{t} X_\tau. \tag{21}$$

Because $X_\tau \ge 0$ for all τ and $X_\tau = 0$ for $\tau < 0$ (by assumption), we may extend the inner sum to the full horizon: for every t, $\sum_{\tau=t-w+1}^{t} X_\tau \le \sum_{\tau=0}^{T-1} X_\tau$. Substituting this into (21) yields the pathwise bound

$$\begin{aligned}\sum_{t=0}^{T-1}\mathbf{1}(E_t)\sum_{k=0}^{w-1} X_{t-k} &\le \sum_{t=0}^{T-1}\mathbf{1}(E_t)\left(\sum_{\tau=0}^{T-1} X_\tau\right) \\ &= N_E\sum_{\tau=0}^{T-1} X_\tau.\end{aligned} \tag{22}$$

Plugging (22) back into (20) gives

$$\sum_{t=0}^{T-1} X_t\mathbf{1}(E_t) \le \frac{N_E}{w}\sum_{\tau=0}^{T-1} X_\tau + \frac{wD_{\max}}{2}N_E.$$

Finally, since $w \ge N_E/\delta$, we have $\frac{N_E}{w} \le \delta$, and since $w = \lceil N_E/\delta\rceil \vee 1$ we also have $w \le N_E/\delta + 1$. Therefore,

$$\sum_{t=0}^{T-1} X_t\mathbf{1}(E_t) \le \delta\sum_{\tau=0}^{T-1} X_\tau + \frac{D_{\max}}{2}\Big(\frac{N_E}{\delta} + 1\Big)N_E.$$

Taking expectations over the sample path completes the proof and yields (18). $\square$

Bounding the UCB failure term (Term II): Apply Lemma 2 with $X_t = Q_{\rm sum}(t)$ and $E_t = \mathcal{E}_t^c$. Choose $\delta_2 := \epsilon/(4S_{\max})$. Also define $N_{\rm err} := \sum_{t=0}^{T-1} \mathbf{1}(\mathcal{E}_t^c)$. Then

$$\begin{aligned} II &\le S_{\max}\delta_2 \sum_{t=0}^{T-1} \mathbb{E}[Q_{\rm sum}(t)] \\ &\quad + S_{\max}\frac{D_{\max}}{2}\Big(\frac{1}{\delta_2}+1\Big)\mathbb{E}[N_{\rm err}^2] \end{aligned}$$

Using Lemma 1, $\mathbb{P}(\mathcal{E}_t^c) \le 2Kt^{-4}$, and the inequality $\mathbb{E}[(\sum_t Z_t)^2] \le (\sum_t \sqrt{\mathbb{E}[Z_t]})^2$ for $Z_t \in \{0,1\}$, we get

$$\mathbb{E}[N_{\rm err}^2] = \mathbb{E}\left[\Big(\sum_{t=0}^{T-1}\mathbf{1}(\mathcal{E}_t^c)\Big)^2\right] \le \left(\sum_{t=0}^{T-1}\sqrt{\mathbb{P}(\mathcal{E}_t^c)}\right)^2$$

Now,

$$\sum_{t=0}^{T-1}\sqrt{\mathbb{P}(\mathcal{E}_t^c)} \le 2 + \sum_{t=2}^{T-1}\sqrt{\mathbb{P}(\mathcal{E}_t^c)} \le 2 + \sum_{t=2}^{\infty}\sqrt{2K}\,t^{-2}.$$

We can write

$$\mathbb{E}[N_{\rm err}^2] \le \left(2+\sqrt{2K}\sum_{t=2}^{\infty}t^{-2}\right)^2 \le \left(2+\sqrt{2K}\,\frac{\pi^2}{6}\right)^2. \tag{23}$$

Hence, we get,

$$II \le \frac{\epsilon}{4}\sum_{t=0}^{T-1}\mathbb{E}[Q_{\rm sum}(t)] + C_2 \tag{24}$$

where

$$C_2 := S_{\max}\cdot\frac{D_{\max}}{2}\Big(\frac{1}{\delta_2}+1\Big)\left(2+\sqrt{2K}\frac{\pi^2}{6}\right)^2. \tag{25}$$

Bounding the exploration term (Term I) We bound $\sum_{t=0}^{T-1}\mathbb{E}[Q_{I(t)}(t)\min\{S_{\max},\,2b_{I(t),J(t)}(t)\}]$ by splitting into *small* and *large* bonus regimes. Let $b_{\rm th} := \epsilon/8$ and write

$$\begin{aligned} &Q_{I(t)}\min\{S_{\max},\,2b_{I(t),J(t)}\} \\ &= Q_{I(t)}\min\{S_{\max},\,2b_{I(t),J(t)}\}\,\mathbf{1}\{b_{I(t),J(t)} < b_{\rm th}\} \\ &\;+ Q_{I(t)}\min\{S_{\max},\,2b_{I(t),J(t)}\}\,\mathbf{1}\{b_{I(t),J(t)} \ge b_{\rm th}\}. \end{aligned}$$

a) Small bonus.: On $\{b_{I(t),J(t)} < \epsilon/8\}$ (define this event as $E_{sb}(t)$) we have $\min\{S_{\max},\,2b_{I(t),J(t)}\} \le 2b_{I(t),J(t)} < \epsilon/4$, hence

$$\begin{aligned} &\sum_{t=0}^{T-1}\mathbb{E}\big[Q_{I(t)}(t)\min\{S_{\max},\,2b_{I(t),J(t)}(t)\}\mathbf{1}\{E_{sb}(t)\}\big] \\ &\le \frac{\epsilon}{4}\sum_{t=0}^{T-1}\mathbb{E}[Q_{\rm sum}(t)]. \end{aligned} \tag{26}$$

b) Large bonus: counting.: Define the event $E_t^{\rm L} := \{b_{I(t),J(t)}(t) \ge \epsilon/8\}$ and let $N_{\rm L} := \sum_{t=0}^{T-1}\mathbf{1}(E_t^{\rm L})$. If at time t the selected arm $k = (I(t),J(t))$ satisfies $b_k(t) \ge \epsilon/8$, then (using $t \le T$)

$$\sqrt{\frac{3S_{\max}^2\ln(\max\{t,2\})}{N_k(t)}} \ge \frac{\epsilon}{8},$$

implying (as $T \ge 2$) $N_k(t) \le \frac{192S_{\max}^2}{\epsilon^2}\ln T$. Therefore, each arm k can trigger $E_t^{\rm L}$ at most $\left(\frac{192S_{\max}^2}{\epsilon^2}\ln T + 1\right)$ times, and summing over K arms yields the deterministic bound

$$N_{\rm L} \le K\left(\frac{192S_{\max}^2}{\epsilon^2}\ln T + 1\right) \le c_{\rm L}\ln T + K, \tag{27}$$

with

$$c_{\rm L} := \frac{192KS_{\max}^2}{\epsilon^2}.$$

c) Large bonus: decoupling.: On $E_t^{\rm L}$, $\min\{S_{\max},\,2b_{I(t),J(t)}(t)\} \le S_{\max}$, hence $Q_{I(t)}\min\{S_{\max},\,2b_{I(t),J(t)}\} \le S_{\max}Q_{\rm sum}(t)$. Apply Lemma 2 with $X_t = Q_{\rm sum}(t)$, $E_t = E_t^{\rm L}$, and choose $\delta_1 := \epsilon/(4S_{\max})$:

$$\sum_{t=0}^{T-1}\mathbb{E}\big[Q_{I(t)}(t)\min\{S_{\max},\,2b_{I(t),J(t)}(t)\}\mathbf{1}(E_t^{\rm L})\big] \tag{28}$$

$$\le S_{\max}\sum_{t=0}^{T-1}\mathbb{E}\big[Q_{\rm sum}(t)\mathbf{1}(E_t^{\rm L})\big]$$

$$\le S_{\max}\delta_1\sum_{t=0}^{T-1}\mathbb{E}[Q_{\rm sum}(t)] \tag{29}$$

$$+\frac{S_{\max}D_{\max}}{2}\Big(\frac{1}{\delta_1}+1\Big)\mathbb{E}[N_{\rm L}^2]$$

$$\le \frac{\epsilon}{4}\sum_{t=0}^{T-1}\mathbb{E}[Q_{\rm sum}(t)] + C_1(\ln T)^2, \tag{30}$$

where we used $\mathbb{E}[N_{\rm L}^2] \le (c_{\rm L}\ln T + K)^2$ from (27) and defined

$$C_1 := S_{\max}\cdot\frac{D_{\max}}{2}\Big(\frac{1}{\delta_1}+1\Big)\cdot(c_{\rm L}+2K)^2. \tag{31}$$

(Here we upper bound $(c_{\rm L}\ln T + K)^2 \le (c_{\rm L}+2K)^2(\ln T)^2$ for all $T \ge 2$.)

Combining (26) and (30) yields

$$I \le \frac{\epsilon}{2}\sum_{t=0}^{T-1}\mathbb{E}[Q_{\rm sum}(t)] + C_1(\ln T)^2. \tag{32}$$

4) Putting Them Together: Substitute (24) and (32) into (17):

$$\begin{aligned} \epsilon\sum_{t=0}^{T-1}\mathbb{E}[Q_{\rm sum}(t)] &\le BT + \left(\frac{\epsilon}{4}\sum_{t=0}^{T-1}\mathbb{E}[Q_{\rm sum}(t)] + C_2\right) \\ &+ \left(\frac{\epsilon}{2}\sum_{t=0}^{T-1}\mathbb{E}[Q_{\rm sum}(t)] + C_1(\ln T)^2\right) + \mathbb{E}[L(0)]. \end{aligned}$$

Rearranging,

$$\frac{\epsilon}{4}\sum_{t=0}^{T-1}\mathbb{E}[Q_{\rm sum}(t)] \le BT + C_1(\ln T)^2 + C_2 + \mathbb{E}[L(0)].$$

Divide by T and multiply by $4/\epsilon$ to obtain (4). Taking $T\to\infty$ yields (6), completing the proof of Theorem 1. □

IV. Experiments

We simulate a slotted single-server wireless system with $N = 10$ users and $M = 10$ rate levels. For each user u and rate index $m \in [1,\cdots,M]$, transmitting at level m serves $R_{u,m} = m+1$ packets if the channel can support that rate.

a) Channel model and mean service rates.: For each user u, we first sample a user-specific channel-state distribution p_u over $\{1,\ldots,M\}$ from a Dirichlet distribution to induce moderate heterogeneity across users. Concretely, we draw

$$p_u \sim \mathrm{Dir}(\alpha),$$

where α is a linearly increasing vector over states; we add a small random jitter to α to avoid degenerate draws. Conditioned on p_u, the channel state process $\{C_u(t)\}_{t\ge 0}$ is i.i.d. over time with $\mathbb{P}(C_u(t) = c) = p_{u,c}$.

We adopt a threshold success model: choosing arm (u,m) at time t yields service $R_{u,m}\mathbf{1}\{C_u(t)\ge m\}$. Thus, the true mean service rate of arm (u,m) is

$$\mu_{u,m} = \mathbb{E}[R_{u,m}\mathbf{1}\{C_u(t)\ge m\}] = R_{u,m}\sum_{c=m}^{M} p_{u,c}.$$

b) Arrivals and load parameter.: Arrivals are i.i.d. over time with $A_u(t)\sim \mathrm{Binomial}(A_{\max}, q_u)$ and $A_{\max} = 3$, so $\lambda_u = \mathbb{E}[A_u(t)] = A_{\max}q_u$. We parameterize the offered load by a single $\rho\in(0,1)$ and set

$$\lambda_u = \rho\cdot\frac{1}{N}\max_{m\in[1,\cdots,M]}\mu_{u,m}, \qquad u\in[1,\cdots,N],$$

with $q_u = \lambda_u/A_{\max}$. In this way, ρ scales each user's offered traffic in proportion to its best achievable mean service rate, and the resulting arrival vector lies strictly inside the capacity region for the range of ρ we consider.

c) Baselines and metric.: We compare Algorithm II-B (learning-based rate adaptation) against an oracle scheduler that knows the true channel statistics $\{\mu_{u,m}\}$, hence implements the MaxWeight decision under the true means $\mu_{u,m}$. Figure 1 reports the running time-average per-user total queue length,

$$\frac{1}{t}\sum_{\tau=0}^{t-1}\frac{1}{N}\sum_{u=1}^{N}Q_u(\tau),$$

averaged over 20 Monte Carlo runs.

We observe that Algorithm II-B closely tracks the oracle in the running time-average total queue length, with the gap diminishing over time. This figure is consistent with the stable behavior predicted by the theory at $\rho = 0.85$.

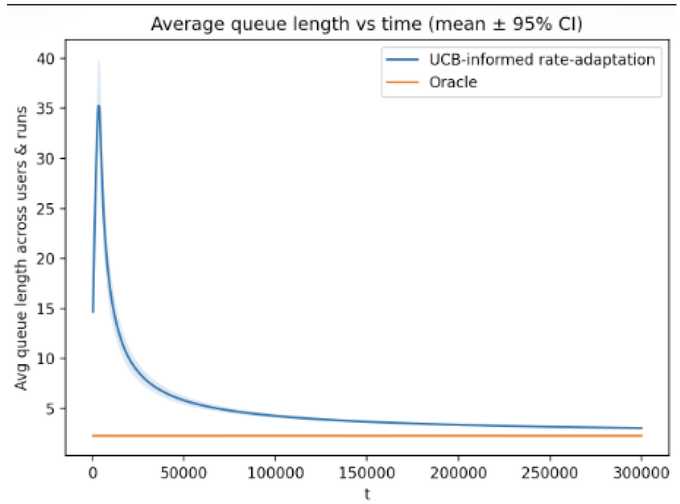

Fig. 1. Running time-average queue length for $N = 10$, $M = 10$ at load $\rho = 0.85$.

V. Conclusion

We considered rate-adaptive scheduling when channel statistics are unknown in a single-server slotted communication network. Treating each (user, rate) pair as a distinct arm, we consider a UCB-informed rate-adaptation algorithm. A decoupling lemma is the key technical tool in the stability analysis based on Lyapunov drift. Establishing finite-time bounds on the expected queue lengths at each slot is an interesting direction for future work.

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